EE6221 CA2 Assignment

State of the Art of General-Purpose Humanoid Robots:
From Whole-Body Control to Vision-Language-Action "Robot Brains"

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Abstract

This report reviews the current state of general-purpose humanoid robots, with a focus on one technology that decides whether they are useful: the control stack that turns camera images and a spoken instruction into coordinated motion of every joint. I compare commercial platforms (Boston Dynamics Atlas, Figure 03, Apptronik Apollo, Unitree G1) and recent research on learned whole-body control and vision-language-action (VLA) models (GR00T N1, π_0, Gemini Robotics 1.5, Helix 02, FAST, HuGe), as well as recent tests of general-purpose frontier models, mainly OpenAI's GPT-6 Astra but also Claude, Gemini and Grok, used as a robot brain. I also compare where each brain can run (on the robot or in the cloud) and how fast it can react, since many applications require offline, on-device operation within a fixed latency budget. I argue that the best current solution is a *hierarchical, learned* architecture in which a slow reasoning model, a fast visuomotor policy, and a very fast whole-body controller are trained to work together, as exemplified by Figure's Helix 02, while Atlas is the most deployment-ready hardware. I then discuss why humanoids are still mostly at the pilot stage while AI agents are already used widely in software, the main open challenges (data, reliability, safety certification, energy, and evaluation), likely future directions, and what I learned.

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1 Introduction and scope

A humanoid robot has a human-like body: two legs, a torso, two arms and, in most current designs, multi-fingered hands. The case for this form is practical rather than aesthetic. Factories, warehouses and homes were built for human bodies, so a robot with the same reach, height and hand shape can in principle use existing tools, shelves and doorways without redesigning the workplace. The same argument explains why humanoids are hard: a body with 50 or more actuated joints that must balance on two small feet is underactuated, has highly coupled dynamics, and must keep doing so while the hands interact with unknown objects.

For many years the hardware was the bottleneck. Today several companies can build reliable humanoid bodies, and the limiting factor has moved to the "brain": how the robot perceives a scene, decides what to do, and converts that decision into stable whole-body motion. This review therefore treats the overall humanoid solution but concentrates on that control stack. It connects directly to EE6221: forward and inverse kinematics of redundant arms, robot dynamics and control, vision sensors and pose estimation, sensor fusion with the Kalman filter, and vision-based control all appear as layers inside a modern humanoid.

Method. I compared systems using public technical reports, peer-reviewed or preprint papers, and official company announcements. Company announcements are treated as *claims*; I note where results lack independent verification. Each solution is judged on five criteria: (i) breadth of tasks it can perform without task-specific programming, (ii) length and reliability of autonomous operation, (iii) data efficiency and ability to transfer across robots, (iv) readiness for real deployment (hardware maturity, safety, integration), and (v) openness and quality of the evidence.

2 Background: the layers of a humanoid control stack

It is useful to describe any humanoid as a stack of loops running at very different rates.

- **L1 Actuation and state estimation** (about 1 kHz). Joint torque or current control, plus estimation of the floating base pose from IMU, joint encoders and foot contact, typically with a Kalman-type filter. The base pose is not measured directly, so this estimate is what every higher layer trusts.
- **L2 Whole-body control (WBC)** (about 50–500 Hz). Given a desired motion of the hands, feet or centre of mass, compute joint commands that achieve it while keeping balance and respecting joint and contact limits.
- **L3 Visuomotor policy** (about 10–200 Hz). Map camera images and proprioception to short-horizon targets such as hand poses, grasp commands, or a reference motion for the WBC.
- **L4 Task reasoning** (about 1–10 Hz or slower). Interpret a language instruction, break it into steps, track progress, and recover from failure.

Classical humanoids built L2 from models: zero-moment-point (ZMP) walking, model predictive control (MPC) of a simplified centroidal model, and a quadratic-program WBC that resolves the redundancy of the kinematic chain (the same null-space idea used for redundant manipulators, but with contact and balance constraints added). L3 and L4 were hand-written state machines. The common thread of the last two years is that *every layer above L1 is being replaced or augmented by learned models*, and the key design question is how those learned layers are divided and connected.

3 Review of the state of the art

3.1 Hardware platforms

Boston Dynamics Atlas (product version). Unveiled at CES on 5 January 2026, the electric Atlas is presented as a production machine rather than a research prototype [1]. The company states 56 degrees of freedom with fully rotational joints, a 50 kg payload, 2.3 m reach, water resistance and operation from −20 to 40 °C. Atlas can walk to a station and swap its own battery, so a low battery does not end a shift. It runs autonomously, by teleoperation or from a tablet, and connects to factory software (MES/WMS) through Boston Dynamics' Orbit platform, which also lets one robot's learned task be copied across a fleet. All 2026 production is committed to Hyundai's Robotics Metaplant Application Center and to Google DeepMind, which is integrating its foundation models into Atlas.

Figure 03. Figure's third-generation robot, announced in October 2025, was redesigned for volume manufacturing and for homes, with a soft textile and foam exterior, a new sensor suite, and hands with palm cameras and tactile sensing [24]. Figure has published few hard specifications; its significance in this review is as the body for the Helix 02 control model (Section 3.3).

Apptronik Apollo is a general-purpose humanoid that Google DeepMind uses as one of the embodiments of Gemini Robotics 1.5 [21].

Unitree G1 is a low-cost humanoid that has become a common research platform; for example, both HuGe [5] and MotionVLA [20] report real-robot experiments on it. Cheap, widely available hardware matters because it lets university groups test learned controllers on a real body instead of only in simulation.

3.2 Learned whole-body control

Reinforcement learning (RL) in GPU-parallel simulation, followed by sim-to-real transfer, is now the dominant way to obtain robust locomotion and whole-body motion for humanoids. The newest trend is the *behaviour foundation model* (BFM): a whole-body controller pre-trained on large motion datasets, often retargeted human motion capture, so that it learns a general repertoire of skills and can be adapted to new tasks zero-shot or with little data. Yuan et al.'s survey, published in IEEE TPAMI in 2026 [4], organises this field by pre-training approach and identifies its open problems: retraining cost for each new scenario, the gap between simulated and real dynamics, and safe adaptation.

FAST (Wang et al., February 2026) [3] is a representative recent method. It keeps a pre-trained tracking policy and adds a small *residual* policy for motions outside the training distribution, with orthogonality (Parseval) and KL constraints so that adapting to a new motion does not erase old skills. It also adds centre-of-mass observations and objectives, which explicitly brings a classical balance quantity back into a learned controller. I find this telling: the best learned controllers are not ignoring model-based insight but feeding it to the network.

3.3 Vision-language-action models

A VLA model starts from a vision-language model pre-trained on internet data and is trained further to output robot actions. The key systems are:

- π_0 (Physical Intelligence, 2024) [23] attaches a flow-matching "action expert" to a vision-language backbone so it can produce smooth, high-rate continuous actions, and trains across many robot types. It established the general recipe that later humanoid models follow.
- **GR00T N1** (NVIDIA, March 2025) [22] is an open humanoid foundation model with a *dual-system* design: a vision-language module ("System 2") interprets the scene and the

instruction, and a diffusion-transformer action module ("System 1") generates fluid motor actions. It is trained on a mixture of real robot teleoperation, human videos and synthetic data, and the weights are public.

- **Gemini Robotics 1.5** (Google DeepMind, October 2025) [21] pairs a VLA with a separate embodied-reasoning model (Gemini Robotics-ER 1.5) that plans and checks progress. The VLA can "think" in natural language before acting, and a *motion transfer* mechanism lets skills learned on one robot run on ALOHA, a bi-arm Franka and the Apollo humanoid without robot-specific post-training.
- **Helix 02** (Figure, early 2026) [2] extends Figure's original upper-body Helix model to the whole body. It adds a learned whole-body controller ("System 0") trained on over 1,000 hours of human motion data and sim-to-real RL, beneath the existing System 1 visuomotor policy and System 2 reasoning model. Figure demonstrated a four-minute, uninterrupted, fully autonomous task of unloading and reloading a dishwasher across a kitchen, which it describes as the longest and most complex autonomous humanoid task to date, and later claimed multi-hour autonomous work shifts. These are company claims without independent evaluation.
- **HuGe** (Liu et al., October 2026) [5] targets the main bottleneck, data. It pre-trains humanoid VLAs on over 2,000 hours of *egocentric* human and robot video. The VLA outputs latent motion tokens for a pre-trained whole-body controller plus hand commands. To bridge the difference between a human and a robot body, it creates pseudo-proprioceptive states by replaying motions in simulation and re-renders the arms in the camera images during fine-tuning. It reports state-of-the-art results on the HumanoidArena benchmark, and on a real Unitree G1 the egocentric data significantly improved success on a pick-and-drop task.

3.4 General-purpose frontier models as the robot brain

A newer idea, which appeared in late 2026, skips robot-specific training entirely. Instead, a general-purpose frontier model, the same kind of model behind today's chatbots and AI agents, is connected to a robot and asked to control it. The most tested example is OpenAI's GPT-6 Astra, which several research groups studied within a few weeks of each other:

- **HomeBody** (Stanford and Caltech, September 2026) [17] runs a Unitree G1 humanoid with Astra. Astra does not output joint commands. It picks a skill (grasp, navigate, open a drawer) and its target through a structured tool call, and the skills run on a local laptop GPU. The robot tidied an unfamiliar kitchen, but the authors note pauses between skills caused by Astra's reasoning latency, extra API cost, and overheating finger servos. It is a project page with code, not a paper with measured results.
- **Astranomy** (NVIDIA, October 2026) [18] tests Astra as a robot policy in simulation without any robot training. Used as an agent with tools, it reached a 74.8% success rate on the RoboLab-120 benchmark, against 28.0% for π_0's successor $\pi_{0.5}$. The cost is speed and money: a single direct call takes about 18.6 s on average, and the agent version uses about 30.9 million tokens (about US$36) per episode.
- **SenseNova's joint study** (October 2026) [19] evaluates Astra on 8 robot configurations and 12 benchmarks, including humanoids. It finds that the best setup uses Astra as a slow "System 2" planner coordinating a fast "System 1" controller such as $\pi_{0.5}$, which it calls "costly but effective". The authors conclude that Astra is not yet practical embodied intelligence, because of cost, execution time and weak recovery from irreversible failures.

These results show that a general model can understand scenes and plan tasks surprisingly well with no robot training. But it is far too slow to close a balance or grasping loop. In every humanoid

example it sits at the top of the stack and calls faster controllers below it. All three sources are preprints or project pages, and I did not find an official technical report from OpenAI describing Astra's design.

Other frontier models. Astra is not the only general model being tried as a robot brain.

- **Gemini.** Google builds its own robotics models on top of Gemini. Gemini Robotics-ER 1.5 is a Gemini model tuned for embodied reasoning and used as the planner above the Gemini Robotics VLA (Section 3.3) [21].
- **Claude.** Robocurve compared Claude Fable 5, Fable 5.1 and Astra on the same pair of 6-DoF YAM arms, with the model sending end-effector poses directly [9]. On placing a block in a bowl (20 trials each), Astra succeeded 19 times, Fable 5.1 8 times and Fable 5 once. On a harder puzzle task, Astra and Fable 5.1 each succeeded only 2 times in 20. The authors note the runs were not interleaved, used different rigs, and were graded by an operator, so this is a small, early comparison. Earlier, Anthropic's Project Fetch (November 2025) used Claude differently: as a coding assistant, it helped a team of four people program a robot dog and finish 7 of 8 tasks, against 6 of 8 for a team without it [11].
- **Grok.** Tesla plans to use xAI's Grok as the conversational "brain" of its Optimus humanoid, while a vision network adapted from Tesla's self-driving software handles perception and movement [12, 13]. Musk has said that for a Grok-level conversation Optimus needs a Wi-Fi or cellular connection to the cloud [14], which is exactly the on-device versus cloud trade-off discussed in Section 4. This is a plan, not a published or tested system.

The pattern is the same for every company: the general model supplies language understanding and planning, and something faster and more specialised moves the body.

3.5 Comparison

Table 1 summarises the solutions against the criteria in the introduction.

4 Deployment constraints: where the brain runs and how fast it must react

Section 3 compared the solutions by what they can do. Before using one, two practical questions also matter. First, what is the *latency budget*, the longest allowed delay between a change in the environment and the robot's response? Second, must the brain run *on-device*, on the robot's own computer, and must it work *offline*, with no network connection? The answers depend on the application, and they decide which models can be used at all.

Table 3 lists what each source states about model size, speed and where it runs. Many companies do not publish these numbers.

Three points follow from the two tables.

1. **Smaller networks run faster, and the pattern is very regular.** Helix 02's 10M-parameter System 0 runs at 1 kHz, Helix's 80M-parameter System 1 at 200 Hz, the 2–7B models at about 7–10 Hz, and the cloud frontier model Astra takes about 18.6 s per decision. Each step up in size buys more reasoning but costs about an order of magnitude in speed.
2. **The latency budget decides which layer a model can be.** Anything that must react within about 100 ms has to be a small network on the robot. Only task planning can tolerate cloud delay, which is why every Astra system in Section 3.4 used Astra as the planner only.

Solution	Architecture	Main strength	Evidence
Atlas (product)	Industrial humanoid; DeepMind models being integrated	Deployment readiness: payload, environment rating, battery self-swap, fleet software	Company specs; first fleets in 2026
GR00T N1	Dual-system VLA (VLM + diffusion action head)	Open weights; real + human-video + synthetic data	Paper; sim and real-robot tests
Gemini Robotics 1.5	VLA + separate embodied-reasoning model	Reasoning before acting; cross-robot motion transfer	Technical report
Helix 02 + Figure 03	Three tiers: S2 reasoning, S1 visuomotor, S0 learned WBC	Single learned system from pixels to whole-body action; long-horizon demo	Company demos only
FAST	Pre-trained WBC + constrained residual adaptation	Fast adaptation without forgetting; CoM-aware balance	Preprint; sim and real
HuGe	VLA → latent motion tokens → WBC	Scales data with egocentric human video	Preprint; benchmark + G1
General frontier models (GPT-6 Astra, Claude Fable, Gemini-ER, Grok)	General model as planner, calling skills or a fast policy	No robot training needed; strong task planning; but slow and costly	Preprints, project pages, small lab tests; Grok only announced

Table 1: Comparison of the reviewed humanoid solutions.

3. **An offline requirement rules out cloud models completely.** Then the top layer must be a model of a few billion parameters that fits on the robot's computer, as in Helix's onboard 7B System 2 or Gemini Robotics On-Device, which is designed to work with intermittent or zero connectivity. For scale, NVIDIA's Jetson AGX Thor, a computer built for humanoids, gives up to 2,070 TFLOPS (sparse FP4) within a 40–130 W power budget and 128 GB of memory [8]. That is enough for several-billion-parameter models, but it uses battery power that would otherwise move the robot.

This gives a second reason for the hierarchical stack. Because the layers are separate, the fast layers can stay on the robot while the slow planner runs in the cloud when a network is available. I suggest that a practical design should also carry a smaller onboard planner as a fallback, so the robot keeps working, more simply, when the connection drops.

5 The best solution and why

In my view the best current solution is the **hierarchical learned stack**: a slow reasoning model (System 2), a fast visuomotor policy (System 1) and a learned whole-body controller (System 0), trained so that the layers share a common interface. The most complete published example is **Figure's Helix 02 on Figure 03**. For a factory buyer today, **Atlas** is the most deployment-ready body, and its pairing with DeepMind's models points to the same architecture. I choose this solution for four reasons.

1. **It matches the physics of the problem.** Balance must be corrected within milliseconds, grasping within tens of milliseconds, and task planning within seconds. One network running at one rate cannot serve all three. The three-tier design gives each timescale its own loop. This is the same principle as the cascaded inner and outer loops we study in control, applied

Situation	Example	Latency budget	Where the brain can run
Balance, fall recovery	Slipping or being pushed	about 1–10 ms	On-device only
Contact and grasp correction	An object slips in the hand	about 10–50 ms	On-device
Reacting to people	A person steps into the robot's path	about 100 ms	On-device
Task planning	"Tidy the kitchen"	about 1–10 s	Cloud is acceptable if a network is available
Offline sites	Factory with no internet or strict data privacy; disaster site	Whole stack	Everything on-device, so smaller models

Table 2: Indicative latency budgets by situation. These are my estimates, based on the loop rates in Section 2, not values from a single source.

to learned components. Section 4 shows that the same split also decides where each layer can run: the fast loops on the robot, and the slow planner on the robot or in the cloud.

2. **It removes the brittle hand-written boundary between walking and manipulation.** In older humanoids, locomotion and arm control were separate controllers joined by a state machine, which is why many demos showed a robot walking, stopping, and only then reaching. Helix 02's System 0 lets the upper layers command whole-body motion directly, so walking while carrying, bending while reaching, and balancing during contact become one behaviour. The dishwasher demonstration, with dozens of sequential loco-manipulation actions and no resets, is the strongest public evidence of long-horizon whole-body autonomy I found.
3. **It scales with data rather than with engineering effort.** Adding a new task means collecting demonstrations, not writing a new controller. Independent research supports this direction: GR00T N1 and HuGe show that human video and synthetic data can supplement scarce robot data, and the BFM literature shows the WBC layer itself can be pre-trained once and reused.
4. **The research community has converged on it.** GR00T N1's dual-system design, Gemini Robotics 1.5's split between reasoning and action, and HuGe's "VLA outputs tokens for a pre-trained WBC" interface are all variants of the same hierarchy. Even the GPT-6 Astra studies (Section 3.4) ended up placing the general model on top as System 2, with faster controllers below it. When independent groups arrive at the same structure, that is good evidence the structure is right, even if the details differ.

I rank Helix 02 first as an integrated system but with a clear qualification: its evidence is company demonstrations, with no public benchmark or paper. If openness and reproducibility are weighted more heavily, GR00T N1 (open weights) and Gemini Robotics 1.5 (detailed technical report) are better references for how the architecture works.

6 Challenges in improving humanoid robots

Why are there so few real applications? LLMs and AI agents are already part of everyday software work, such as writing code and answering customers. Humanoids, which use the same kind of models as their "brain", are still mostly at the pilot stage. The public record is small. Agility Robotics and the logistics company GXO signed what they called the first humanoid Robot-as-a-Service contract in 2024 [16]. Figure reports about 740 robots operating by the end of June 2026, which is a company figure [15]. Tesla is still installing its first Optimus production lines, and

System	Parameters	Rate / delay	Where it runs	Training data
Helix [6]	S2: 7B; S1: 80M	S2: 7–9 Hz; S1: 200 Hz	Fully onboard, embedded low-power GPUs	About 500 h teleoperation
Helix 02 [2]	S0: 10M; others not disclosed	S1: 200 Hz; S0: 1 kHz	Not stated	Over 1,000 h human motion (for S0)
GR00T N1 [22]	2.2B (1.34B in the VLM)	VLM: 10 Hz; action head: 120 Hz; 63.9 ms per 16-action chunk	Timed on an L40 data-centre GPU	Real, human-video and synthetic
π_0 [23]	3.3B	Up to 50 Hz	Mobile robot: off-board over Wi-Fi	About 10,000 h demonstrations
Gemini Robotics On-Device [7]	Not disclosed	"Low-latency", no figure	On the robot; works with no network	Adapts from 50–100 demos
HuGe [5]	Qwen-based VLM; total not stated	50 Hz action timeline	Not stated	Over 2,000 h egocentric video
GPT-6 Astra [18]	Not disclosed	About 18.6 s per call	Cloud API	No robot-specific training

Table 3: Size, speed and location of the reviewed robot brains, as stated in each source.

analysts expect external sales no earlier than the second half of 2027 [13].

In my view the gap comes from four differences between software agents and robots:

- **Mistakes cost more.** A wrong line of code can be reviewed, tested and undone. A wrong physical action can break equipment or hurt someone, and it cannot be undone.
- **Each copy costs money.** A software agent can be copied to a million users almost for free. Every robot is hardware that must be built, charged, maintained and insured.
- **The bar is a working alternative.** A humanoid must beat a fixed automation cell or a human worker on total cost and reliability, and both already work well.
- **There is less data and less certainty.** Robot data is scarce, and there is no safety standard yet for certifying the robot.

The challenges below are the technical side of this gap. Note that LLM progress does reach robots quickly, but only at the top of the stack, as the planner. The body, the fast control loops and safety are what still limit commercial use.

1. **Data scarcity and embodiment gap.** Language models learned from trillions of words; humanoid policies have thousands of hours of demonstrations. Teleoperation is slow and expensive, and human video does not contain the robot's joint torques or contact forces. HuGe's need for simulation replay and arm re-rendering shows how much work is required just to make human data usable.
2. **Reliability in the long tail.** A demo succeeding once is very different from 99.9% success over an eight-hour shift. Learned policies can fail in surprising ways when lighting, objects or clutter differ from training data, and these failures are hard to predict in advance.
3. **Safety and certification.** A humanoid that falls when power is lost is a new hazard class. Existing standards do not cover it: ISO 10218 does not address mobility, ISO 3691-4

excludes these robots, and ISO 13482 is not for industrial sites. The dedicated standard for dynamically stable industrial mobile robots, ISO 25785-1, is still in development [25]. It is also unclear how to certify a neural network controller whose behaviour cannot be fully specified in advance. General models make this harder. In Robocurve's RoboHarm test (September 2026), Claude Fable 5.1, GPT-6 Astra and the robot model MolmoAct2 controlled real arms and were each given 5 instructions a safe robot should refuse, 20 times each [10]. Fable refused 20 of 100, all on one task (stabbing a doll); Astra refused 2; MolmoAct2 refused none. Fable and Astra each pushed a metal screwdriver into a toaster 6 and 7 times out of 20. The chatbot safety training of these models does not carry over to physical actions, so a separate safety layer on the robot is still needed.

4. **Hands and touch.** Dexterous manipulation still lags far behind locomotion. Multi-fingered hands are fragile and costly, and tactile sensing is only now being integrated into learned policies (for example the palm cameras and tactile sensors of Figure 03).
5. **Energy, compute and cost.** Running large VLA models on board at high rates competes with the battery used for motion. Battery swapping, as on Atlas, works around limited runtime rather than solving it. Unit cost must also fall far enough for a humanoid to beat a fixed automation cell or a human worker on total cost.
6. **Evaluation.** Much of the most impressive work is shown only in company videos. Without shared benchmarks and independent testing, it is difficult to compare systems or know which advance actually caused an improvement.

7 Future directions

- **Scaling data from humans and simulation.** Egocentric human video (as in HuGe), large synthetic datasets, and world models that predict future observations should reduce dependence on teleoperation.
- **Learning from deployment.** Fleets connected through software such as Orbit can pool experience, so a correction learned by one robot improves all of them. Reinforcement learning on real deployment data is a natural next step after imitation learning.
- **Combining learning with model-based guarantees.** FAST's centre-of-mass objectives suggest a hybrid future: learned policies for flexibility, wrapped by model-based safety filters (for example balance or collision constraints solved as a small optimisation) that can be verified and certified.
- **Richer sensing.** Tactile skins, in-hand cameras and force sensing fused with vision, using the estimation tools of this course, will be needed for reliable contact-rich work.
- **Standards and benchmarks.** Completing ISO 25785-1 and creating shared simulation and real-world benchmarks (HumanoidArena is an early example) will move the field from demos to measurable progress.
- **Efficient on-board models.** Smaller, distilled VLAs and dedicated accelerators will cut power and latency so that more of the stack runs locally without a network connection.

8 What I learned from this review

[DRAFT — rewrite in your own words and keep only what is true for you.]

- The classical material in EE6221 has not been replaced by AI; it has moved to a different place in the stack. Kinematics and dynamics now appear as the structure of the whole-body

controller, as balance quantities such as the centre of mass fed into a learned policy, and as the retargeting that maps human motion onto a robot's joints. State estimation of the floating base is still a filtering problem.

- The most important design decision in a modern humanoid is how to split the control problem by timescale. This is a familiar control idea (cascaded loops), now applied to neural networks.
- Data, not algorithms, is currently the binding constraint, which is why so much recent research is really about finding new sources of data.
- I learned to separate evidence types: a peer-reviewed paper, a preprint and a company video support claims to very different degrees. [Add one or two specific things that surprised you.]

9 Effect of this assignment on my understanding and interest

[This section must be your own honest answer; the points below are prompts only.]

- **Understanding of robotics and intelligent systems:** [Did it increase? Which link between the lectures (kinematics, dynamics and control, vision sensors, Kalman filtering, vision-based control) and real humanoids became clearer?]
- **Interest in a career in robotics and intelligent systems:** [Did it increase? Which part of the stack (control, perception, learning, safety) would you want to work on, and why?]

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